Operations Research, Central Exercise 3

Revised Simplex & Sensitivity Analysis

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Revised Simplex

Matrix Formulation of an LP

$$\max \; c^T x \qquad \text{s.t. } Ax = b, \; x \geq 0$$

- Partition columns of constraint matrix:

$$A = [B \; N]$$

- Partition variables and coefficients vectors:

$$x = \begin{pmatrix} x_B \\ x_N \end{pmatrix}, \qquad c = \begin{pmatrix} c_B \\ c_N \end{pmatrix}$$

- System becomes:

$$Bx_B + Nx_N = b$$

Basis Transformation

- Multiply constraint by B^{-1}:

$$B^{-1}Bx_B + B^{-1}Nx_N = B^{-1}b$$

$$\Rightarrow \; x_B + B^{-1}Nx_N = B^{-1}b$$

- Define:

$$N' := B^{-1}N, \qquad b' := B^{-1}b$$

- Then:

$$x_B = b' - N'x_N$$

Transformed Objective Function

- Objective:

$$z = c_B^T x_B + c_N^T x_N$$

- Substitute:

$$x_B = b' - N' x_N$$

- Then:

$$z = c_B^T b' + \left(c_N^T - c_B^T N' \right) x_N$$

- Rearranging:

$$z - \left(c_N^T - c_B^T N' \right) x_N = c_B^T b'$$

- Define:

$$c_N'^T := - \left(c_N - N'^T c_B \right)^T$$

- Final form:

$$z - c_N'^T x_N = c_B^T b'$$

Final Form

- Transformed system:

$$x_B = b' - N' x_N$$

- Objective in tableau form:

$$z - c_N'^T x_N = c_B^T b'$$

- Definitions:

$$b' = B^{-1} b, \qquad N' = B^{-1} N$$

$$c_N'^T = -\left(c_N - N'^T c_B \right)^T$$

- Current basic feasible solution ($x_N = 0$):

$$x_B = b', \qquad z = c_B^T b'$$

Comparison to Simplex Tableau

Basic Variable	Coefficient							Solution
	z	x_{B_1}	…	x_{B_m}	x_{N_1}	…	$x_{N_{n-m}}$	
z	1	0		0	$c_N'^T = -\left(c_N - N'^T c_B\right)^T$			$c_B^T B^{-1} b$
x_{B_1}		1						
…			…			$N' = B^{-1}N$		$b' = B^{-1}b$
x_{B_m}				1				

Important: Understand how each entry in the tableau is derived and what it represents (see, e.g., Exercise S3.3 from Retake 2025).

Feasibility and Optimality Conditions

- Transformed system:

$$x_B = b' - N' x_N, \qquad z - c_N'^T x_N = c_B^T b'$$

- **Feasibility:**

$$x \geq 0 \iff b' = B^{-1} b \geq 0$$

- **Optimality (max problem):**

$$c_N' \geq 0 \ \ \forall j \in N$$

- **Current solution:**

$$x = \begin{pmatrix} b' \\ 0 \end{pmatrix}, \qquad z = c_B^T b'$$

- **Interpretation:**
  - Feasible $\Rightarrow$ basic variables non-negative
  - Optimal $\Rightarrow$ no improving non-basic direction

Revised Simplex Algorithm

1. **Initialization:**
   - Find an initial basis B (basic variables)
2. **Repeat:**
   - Compute B^{-1}
   - Compute:
     $$N' = B^{-1}N$$
   - Compute reduced costs:
     $$c'_N = -\left(c_N - N'^T c_B\right)$$
   - **Entering variable:**
     - Choose $j \in \arg\min_{k \in N} c'_k \quad$ with $c'_j < 0$
   - **Leaving variable:**
     - Use N', x_B and ratio test to maintain feasibility
   - **Update:**
     - Perform basis swap
     - Update B, N, c_B, c_N
3. **Until:**
   - $c'_N \geq 0$ (optimal solution), or
   - problem is unbounded:
     $\exists j \in N$ with $c'_j < 0 \quad$ and $\quad d := B^{-1}a_j \leq 0$

Exercise D3.1: Revised Simplex

Problem:

$$\begin{aligned} \max\ & 12x_1 + 20x_2 + 16x_3 + 12x_4 \\ & 4x_1 + 8x_2 + 2x_3 \leq 400 \\ \text{s.t.}\ & 2x_1 + 5x_2 + 10x_3 + 4x_4 \leq 400 \\ & x_i \geq 0 \end{aligned}$$

Start with the feasible basis (x_1, x_6).

Exercise D3.1: Revised Simplex

Problem:

$$\max\ 12x_1 + 20x_2 + 16x_3 + 12x_4$$

$$\begin{aligned} 4x_1 + 8x_2 + 2x_3 &\leq 400 \\ \text{s.t.}\ \ 2x_1 + 5x_2 + 10x_3 + 4x_4 &\leq 400 \\ x_i &\geq 0 \end{aligned}$$

Start with the feasible basis (x_1, x_6).

Canonical Form:

$$\max\ 12x_1 + 20x_2 + 16x_3 + 12x_4$$

$$\begin{aligned} 4x_1 + 8x_2 + 2x_3 + x_5 &= 400 \\ \text{s.t.}\ \ 2x_1 + 5x_2 + 10x_3 + 4x_4 + x_6 &= 400 \\ x_i &\geq 0 \end{aligned}$$

$$A = \begin{pmatrix} 4 & 8 & 2 & 0 & 1 & 0 \\ 2 & 5 & 10 & 4 & 0 & 1 \end{pmatrix}, \quad b = \begin{pmatrix} 400 \\ 400 \end{pmatrix}$$

$$c = \begin{pmatrix} 12 & 20 & 16 & 12 & 0 & 0 \end{pmatrix}^T$$

Iteration 1

$$A = \begin{pmatrix} 4 & 8 & 2 & 0 & 1 & 0 \\ 2 & 5 & 10 & 4 & 0 & 1 \end{pmatrix}$$

Basis:

$$x_B = (x_1, x_6)^T, \quad x_N = (x_2, x_3, x_4, x_5)^T$$

$$B = \begin{pmatrix} 4 & 0 \\ 2 & 1 \end{pmatrix}, \quad N = \begin{pmatrix} 8 & 2 & 0 & 1 \\ 5 & 10 & 4 & 0 \end{pmatrix}$$

Check: Valid basis? This is not the canonical basis $B = I$!

Iteration 1

$$A = \begin{pmatrix} 4 & 8 & 2 & 0 & 1 & 0 \\ 2 & 5 & 10 & 4 & 0 & 1 \end{pmatrix}$$

Basis:

$$x_B = (x_1, x_6)^T, \quad x_N = (x_2, x_3, x_4, x_5)^T$$

$$B = \begin{pmatrix} 4 & 0 \\ 2 & 1 \end{pmatrix}, \quad N = \begin{pmatrix} 8 & 2 & 0 & 1 \\ 5 & 10 & 4 & 0 \end{pmatrix}$$

Check: Valid basis? This is not the canonical basis $B = I$!
Yes, because $x_B = B^{-1}b \geq 0$.

Iteration 1: Computing B^{-1}

$$B = \begin{pmatrix} 4 & 0 \\ 2 & 1 \end{pmatrix}$$

$$[B \mid I] = \begin{pmatrix} 4 & 0 & 1 & 0 \\ 2 & 1 & 0 & 1 \end{pmatrix}$$

- $R_1 \leftarrow \frac{1}{4} R_1$
- $R_2 \leftarrow R_2 - 2R_1$

$$B^{-1} = \begin{pmatrix} \frac{1}{4} & 0 \\ -\frac{1}{2} & 1 \end{pmatrix}$$

Iteration 1: Computations

$$N' = B^{-1}N = \begin{pmatrix} 2 & \frac{1}{2} & 0 & \frac{1}{4} \\ 1 & 9 & 4 & -\frac{1}{2} \end{pmatrix}$$

$$c = \begin{pmatrix} 12 & 20 & 16 & 12 & 0 & 0 \end{pmatrix}^T$$

$$c_B = \begin{pmatrix} 12 \\ 0 \end{pmatrix}, \quad c_N = \begin{pmatrix} 20 & 16 & 12 & 0 \end{pmatrix}^T$$

$$c'_N = -(c_N - N'^T c_B) = \begin{pmatrix} 4 \\ -10 \\ -12 \\ 3 \end{pmatrix}$$

$$x_B = B^{-1}b = \begin{pmatrix} 100 \\ 200 \end{pmatrix}$$

Iteration 1: Pivot

- Reduced costs:

$$c'_N = (4, -10, -12, 3)^T$$

- **Entering variable:** Most negative reduced cost: $c'_4 = -12$. Therefore: x_4 enters.
- Column of x_4 in N':

$$N' = \begin{pmatrix} 2 & \frac{1}{2} & 0 & \frac{1}{4} \\ 1 & 9 & 4 & -\frac{1}{2} \end{pmatrix}, \quad N'_4 = \begin{pmatrix} 0 \\ 4 \end{pmatrix}$$

- Current basic solution:

$$x_B = \begin{pmatrix} x_1 \\ x_6 \end{pmatrix} = \begin{pmatrix} 100 \\ 200 \end{pmatrix}$$

- Ratio test: x_6 leaves
- **Pivot:** x_4 enters, x_6 leaves

Iteration 2

$$A = \begin{pmatrix} 4 & 8 & 2 & 0 & 1 & 0 \\ 2 & 5 & 10 & 4 & 0 & 1 \end{pmatrix}$$

New basis:

$$x_B = (x_1, x_4)^T, \quad x_N = (x_2, x_3, x_5, x_6)^T$$

$$B = \begin{pmatrix} 4 & 0 \\ 2 & 4 \end{pmatrix}, \quad N = \begin{pmatrix} 8 & 2 & 1 & 0 \\ 5 & 10 & 0 & 1 \end{pmatrix}$$

$$B^{-1} = \begin{pmatrix} \frac{1}{4} & 0 \\ -\frac{1}{8} & \frac{1}{4} \end{pmatrix}$$

Iteration 2: Computations

$$N' = B^{-1}N = \begin{pmatrix} 2 & \frac{1}{2} & \frac{1}{4} & 0 \\ \frac{1}{4} & \frac{9}{4} & -\frac{1}{8} & \frac{1}{4} \end{pmatrix}$$

$$c = \begin{pmatrix} 12 & 20 & 16 & 12 & 0 & 0 \end{pmatrix}^T$$

$$c_B = \begin{pmatrix} 12 \\ 12 \end{pmatrix}, \quad c_N = \begin{pmatrix} 20 & 16 & 0 & 0 \end{pmatrix}^T$$

$$c'_N = -(c_N - N'^T c_B) = \begin{pmatrix} 7 \\ 17 \\ \frac{3}{2} \\ 3 \end{pmatrix}$$

$$x_B = B^{-1}b = \begin{pmatrix} 100 \\ 50 \end{pmatrix}$$

Optimal Solution

- All reduced costs:

$$c'_N \geq 0 \quad \text{(no improving direction)}$$

- Optimal basic solution:

$$x_1 = 100, \quad x_4 = 50$$

- Objective value:

$$Z = c_B^T B^{-1} b = 1800$$

Sensitivity Analysis

Sensitivity Analysis: Goal

- Study how the optimal solution changes when LP parameters change.
- Main questions:
  - What happens if the right-hand side b changes?
  - What happens if the objective coefficients c change?
  - How can we evaluate the value of resources?
  - How can we reoptimize efficiently after adding a new variable?
- Work with the current optimal basis B.

Recap: Simplex Tableau in Matrix Notation

Basic Variable	Coefficient							Solution
	z	x_{B_1}	…	x_{B_m}	x_{N_1}	…	$x_{N_{n-m}}$	
z	1	0		0	$c_N'^T = -(c_N - N'^T c_B)^T$			$c_B^T B^{-1} b$
x_{B_1}		1			$N' = B^{-1}N$			$b' = B^{-1}b$
…			…					
x_{B_m}				1				

Current basic solution:

$$x_N = 0 \quad \Rightarrow \quad x_B = b' = B^{-1} b.$$

Feasibility: basic solution is non-negative

$$b' = B^{-1} b \geq 0.$$

Optimality for maximization: reduced costs are non-negative

$$c_N' = -\left(c_N - N'^T c_B\right) \geq 0$$

Change in the Right-Hand Side

Let the RHS change from b to

$$b + \Delta b.$$

The basis B stays fixed, so

$$x_B^{\text{new}} = B^{-1}(b + \Delta b) = b' + B^{-1}\Delta b.$$

- Reduced costs c'_N do not change if only b changes $\longleftrightarrow$ optimality not affected.
- Only feasibility may be lost.

Condition for feasibility:

$$b' + B^{-1}\Delta b \geq 0.$$

Shadow Prices

The objective value at the current basis is

$$z^* = c_B^T x_B = c_B^T B^{-1} b.$$

If b changes to $b + \Delta b$, then

$$z^{\text{new}} = z^* + c_B^T B^{-1} \Delta b.$$

Define shadow prices:

$$y^T := c_B^T B^{-1}.$$

$y_i =$ marginal increase in z from one extra unit of resource i.

- y is determined by the current basis B.

Validity: Only valid while the current basis B remains optimal.

Shadow Prices and Reduced Costs

Given the shadow prices:

$$y^T = c_B^T B^{-1}$$

The **reduced costs** are defined as:

$$c_j' = -\left(c_j - a_j^T y\right), \quad j \in N.$$

Interpretation:

- c_j = direct profit of variable x_j
- $a_j^T y$ = value of resources used (at shadow prices)

Economic meaning:

$$c_j - a_j^T y = \text{net profit of introducing } x_j.$$

- If $c_j' \geq 0$: not profitable $\Rightarrow$ keeping $x_j = 0$ is optimal
- If $c_j' < 0$: profitable $\Rightarrow$ enter basis

Shadow Prices in the Simplex Tableau

Reduced costs:

$$c'_j = -(c_j - a_j^T y), \quad \text{with } y^T = c_B^T B^{-1}$$

For slack variables s_i:

$$c_{s_i} = 0, \quad a_{s_i} = e_i \;\Rightarrow\; c'_{s_i} = y_i$$

- Entries under slack variables in the objective row = shadow prices

Case	**Slack s_i**	**Constraint**	**Shadow Price y_i**
Non-binding	$s_i > 0$	Not tight (unused resource)	$y_i = 0$
Binding	$s_i = 0$	Tight (fully used resource)	$y_i > 0$

Complementary slackness condition (next lecture):

$$y_i \cdot s_i = 0$$

Change in the Objective Coefficients

Now keep A, b fixed, but change the objective coefficients c.

If the basis remains the same, then

$$x_B = B^{-1}b$$

does not change.
But the reduced costs do change:

$$c'_N = -(c_N - N'^T c_B).$$

If c_B or c_N changes by Δ_d, recompute c'_N, i.e., $c'_N(\Delta_d)$.

Condition for the current basis to remain optimal:

$$c'_N(\Delta_d) \geq 0$$

Reoptimization After Adding a New Variable

Suppose a new variable x_{new} with column a_{new} and objective coefficient c_{new} is added.

Do not solve the LP from scratch!

Instead: Use the current optimal basis B and compute:

$$\bar{a} = B^{-1} a_{new}.$$

The reduced cost of the new variable is

$$c'_{new} = -\left(c_{new} - \bar{a}^T c_B\right).$$

- If $c'_{new} \geq 0$, the current basis remains optimal.
- If $c'_{new} < 0$, the new variable should enter the basis.

Adding a new constraint is harder because it may destroy feasibility!

Exercise D3.2, a): Waldgeist Distillery

Resources	Himmelrot	Nachtblau	Capacity
Raspberry	2	1	20
Blueberry	2	3	18
Strawberry	1	1/2	8
Unit price	15	20	
Unit cost	10	16	

$$
\begin{aligned}
\max\ & 5x_1 + 4x_2 \\
\text{s.t. } & 2x_1 + x_2 \leq 20 \\
& 2x_1 + 3x_2 \leq 18 \\
& x_1 + 1/2x_2 \leq 8 \\
& x \in \mathbb{R}^2_{\geq 0}
\end{aligned}
$$

Exercise D3.2, a): Waldgeist Distillery

Decision variables:

x_1 = Himmelrot bottles produced, x_2 = Nachtblau bottles produced

Maximize profit:

$$15x_1 + 20x_2 - (10x_1 + 16x_2) = 5x_1 + 4x_2$$

$$\max\ 5x_1 + 4x_2$$

Capacity constraints:

$$\begin{aligned} 2x_1 + x_2 &\leq 20 \quad \text{(raspberries)} \\ 2x_1 + 3x_2 &\leq 18 \quad \text{(blueberries)} \\ x_1 + \frac{1}{2}x_2 &\leq 8 \quad \text{(strawberries)} \end{aligned}$$

Non-negativity constraint:

$$x_1, x_2 \geq 0$$

b) Final Simplex Tableau

	x_1	x_2	x_3	x_4	x_5	Res.
Z			0	0.75	3.5	
			1	0	-2	
x_2			0	0.5	-1	
x_1			0	-0.25	1.5	7.5

b) Final Simplex Tableau

	x_1	x_2	x_3	x_4	x_5	Res.
Z	0	0	0	0.75	3.5	41.5
x_3	0	0	1	0	-2	4
x_2	0	1	0	0.5	-1	1
x_1	1	0	0	-0.25	1.5	7.5

What is the economic interpretation of x_3, x_4, x_5?

b) Final Simplex Tableau

	x_1	x_2	x_3	x_4	x_5	Res.
Z	0	0	0	0.75	3.5	41.5
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x_2	0	1	0	0.5	-1	1
x_1	1	0	0	-0.25	1.5	7.5

What is the economic interpretation of x_3, x_4, x_5?

- x_3: unused kg of rasberries
- x_4: unused kg of blueberries
- x_5: unused kg of strawberries

c) Shadow Price of Raspberries

What is the maximum amount that Steffen should pay for 1kg of additional raspberries? Justify your answer.

	x_1	x_2	x_3	x_4	x_5	Res.
Z	0	0	0	0.75	3.5	41.5
x_3	0	0	1	0	-2	4
x_2	0	1	0	0.5	-1	1
x_1	1	0	0	-0.25	1.5	7.5

c) Shadow Price of Raspberries

What is the maximum amount that Steffen should pay for 1kg of additional raspberries? Justify your answer.

	x_1	x_2	x_3	x_4	x_5	Res.
Z	0	0	0	0.75	3.5	41.5
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x_2	0	1	0	0.5	-1	1
x_1	1	0	0	-0.25	1.5	7.5

The shadow price of an additional unit of raspberries is 0. (There are still 4 units of raspberries left.) Therefore, Steffen should not spend anything on additional raspberries.

d) Selling Price Limits for Nachtblau - Analyze Change in Objective Coefficients

Calculate the limits for the selling price of Nachtblau within which the optimal basis does not change. Explain in detail how the optimal production schedule and resource stocks change, if the selling price violates these limits.

Recap: Objective Coefficient Sensitivity

Change one objective coefficient:

$$c_j \to c_j + \Delta$$

- A, b stay fixed $\Rightarrow$ feasibility does not change.
- Only reduced costs change.
- Same basis remains optimal iff:

$$c'_N(\Delta) \geq 0$$

If the changed variable is basic:

$$c_B \to c_B + \Delta e_i$$

$$c'_N(\Delta) = c'_N + \Delta({N'}^T e_i)$$

Tableau rule: Modify the Z-row, eliminate the basic variable again, then require all reduced costs ≥ 0.

d) Selling Price Limits for Nachtblau - Analyze Change in Objective Coefficients

Let the price of Nachtblau change by Δ.

	x_1	x_2	x_3	x_4	x_5	RHS
Z	0	$-\Delta$	0	0.75	3.5	41.5
x_3	0	0	1	0	-2	4
x_2	0	1	0	0.5	-1	1
x_1	1	0	0	-0.25	1.5	7.5

d) Selling Price Limits for Nachtblau - Analyze Change in Objective Coefficients

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Z	0	$-\Delta$	0	0.75	3.5	41.5
x_3	0	0	1	0	-2	4
x_2	0	1	0	0.5	-1	1
x_1	1	0	0	-0.25	1.5	7.5

Transform such that the entry of x_2 in Z row is 0 (x_2 is basic):

	x_1	x_2	x_3	x_4	x_5	RHS
Z	0	0	0	$0.75 + \frac{\Delta}{2}$	$3.5 - \Delta$	$41.5 + \Delta$

d) Selling Price Limits for Nachtblau - Analyze Change in Objective Coefficients

Let the price of Nachtblau change by Δ.

	x_1	x_2	x_3	x_4	x_5	RHS
Z	0	$-\Delta$	0	0.75	3.5	41.5
x_3	0	0	1	0	-2	4
x_2	0	1	0	0.5	-1	1
x_1	1	0	0	-0.25	1.5	7.5

Transform such that the entry of x_2 in Z row is 0 (x_2 is basic):

	x_1	x_2	x_3	x_4	x_5	RHS
Z	0	0	0	$0.75 + \frac{\Delta}{2}$	$3.5 - \Delta$	$41.5 + \Delta$

For the current basis to remain optimal, it must hold $c'_N \geq 0$, i.e.,

$$0.75 + \frac{\Delta}{2} \geq 0 \text{ and } 3.5 - \Delta \geq 0 \iff -1.5 \leq \Delta \leq 3.5$$

d) Selling Price Limits for Nachtblau - Analyze Change in Objective Coefficients

Let the price of Nachtblau change by Δ.

	x_1	x_2	x_3	x_4	x_5	RHS
Z	0	$-\Delta$	0	0.75	3.5	41.5
x_3	0	0	1	0	-2	4
x_2	0	1	0	0.5	-1	1
x_1	1	0	0	-0.25	1.5	7.5

Transform such that the entry of x_2 in Z row is 0 (x_2 is basic):

	x_1	x_2	x_3	x_4	x_5	RHS
Z	0	0	0	$0.75 + \frac{\Delta}{2}$	$3.5 - \Delta$	$41.5 + \Delta$

For the current basis to remain optimal, it must hold $c'_N \geq 0$, i.e.,

$$0.75 + \frac{\Delta}{2} \geq 0 \text{ and } 3.5 - \Delta \geq 0 \iff -1.5 \leq \Delta \leq 3.5$$

$$p_N \in [20 - 1.5,\ 20 + 3.5] = [18.5, 23.5]$$

d) Selling Price Limits for Nachtblau

Price can vary within $p_N \in [18.5, 23.5]$ for $\Delta \in [-1.5, 3.5]$ without basis change.

If the price of Nachtblau **falls below** 18.5**, i.e., if** $\Delta < -1.5$, then:

$$0.75 + \frac{\Delta}{2} < 0$$

	x_1	x_2	x_3	x_4	x_5	RHS
Z	0	0	0	$0.75 + \frac{\Delta}{2}$	$3.5 - \Delta$	$41.5 + \Delta$
x_3	0	0	1	0	-2	4
x_2	0	1	0	0.5	-1	1
x_1	1	0	0	-0.25	1.5	7.5

So the current basis is no longer optimal.

Basis change: x_4 enters, x_2 leaves

New optimal interpretation: produce 8 bottles Himmelrot, 0 Nachtblau; 4 raspberries, 2 blueberries, 0 strawberries left.

d) Selling Price Limits for Nachtblau

Price can vary within $p_N \in [18.5, 23.5]$ for $\Delta \in [-1.5, 3.5]$ without basis change.
If the price of Nachtblau **falls above** 23.5**, i.e., if** $\Delta > 3.5$, then:

$$3.5 - \Delta < 0$$

	x_1	x_2	x_3	x_4	x_5	RHS
Z	0	0	0	$0.75 + \frac{\Delta}{2}$	$3.5 - \Delta$	$41.5 + \Delta$
x_3	0	0	1	0	-2	4
x_2	0	1	0	0.5	-1	1
x_1	1	0	0	-0.25	1.5	7.5

So the current basis is no longer optimal.

Basis change: x_5 enters, x_1 leaves

New optimal interpretation: produce 0 bottles Himmelrot, 6 Nachtblau; 14 raspberries, 0 blueberries, 5 strawberries left.

e) Supply of blueberries/strawberries varies - Analyze Change in RHS

Within what limits can the blueberry/strawberry capacity move so that both products can continue to be produced?

Recap: RHS Sensitivity Analysis

Change in RHS:

$$b \to b + \Delta b$$

Updated basic solution:

$$x_B^{new} = B^{-1}(b + \Delta b) = b' + B^{-1}\Delta b$$

- Reduced costs c'_N do **not** change
- Only feasibility may be lost

For a single constraint i:

- Change RHS by Δ_i
- Then:

$$x_B^{new} = b' + \Delta_i \cdot (B^{-1}e_i)$$

- In the tableau: use the column of the corresponding slack variable x_i

Condition for same basis:

$$x_B^{new} \geq 0$$

e) (i) Blueberry Capacity Allowed Variation - SKIP

Let Δ_B change the RHS of the blueberry constraint.
Using the final tableau:

	x_1	x_2	x_3	x_4	x_5	RHS
Z	0	0	0	0.75	3.5	41.5
x_3	0	0	1	0	-2	4
x_2	0	1	0	0.5	-1	1
x_1	1	0	0	-0.25	1.5	7.5

Updated solution becomes $x_2 = 1 + 0.5\Delta_B$ and $x_1 = 7.5 - 0.25\Delta_B$
Both products produced iff

$$x_2 = 1 + 0.5\Delta_B > 0, \quad x_1 = 7.5 - 0.25\Delta_B > 0$$

Thus:

$$-2 < \Delta_B < 30$$

Since original blueberry capacity is 18:

$$B_{\text{cap}} \in (18 - 2,\ 18 + 30) = (16, 48)$$

e) (ii) Strawberry Capacity Allowed Variation

Let Δ_S change the RHS of the strawberry constraint.
Using the final tableau:

	x_1	x_2	x_3	x_4	x_5	RHS
Z	0	0	0	0.75	3.5	41.5
x_3	0	0	1	0	-2	4
x_2	0	1	0	0.5	-1	1
x_1	1	0	0	-0.25	1.5	7.5

Updated solution becomes

$$x_3 = 4 - 2\Delta_S, \quad x_2 = 1 - \Delta_S, \quad x_1 = 7.5 + 1.5\Delta_S$$

Feasibility and producing both products require:

$$4 - 2\Delta_S \geq 0, \quad 1 - \Delta_S > 0, \quad 7.5 + 1.5\Delta_S > 0$$

Thus:

$$-5 < \Delta_S < 1$$

$$S_{\text{cap}} \in (8 - 5,\ 8 + 1) = (3, 9)$$

e) (iii) Strawberry Capacity Too Low ($\Delta_S = -5$)

Step 1: Check feasibility

$$x_1 = 7.5 + 1.5(-5) = 0 \quad \Rightarrow \quad x_1 \text{ leaves the basis}$$

Step 2: Restore optimality

Let t be the new coefficient of x_1 in the Z-row:

	x_1	x_2	x_3	x_4	x_5
$Z(t)$	t	0	0	$0.75 - 0.25t$	$3.5 + 1.5t$
x_1	1	0	0	-0.25	1.5

Step 3: Optimality condition: $0.75 - 0.25t \geq 0 \quad \Rightarrow \quad t \leq 3$

Step 4: Pivot At $t = 3$: x_4 enters, $\quad x_1$ leaves

Result (new shadow prices / reduced costs): $y = (0,\ 0,\ 8)$

$\Rightarrow$ Strawberry shadow price increases from 3.5 to 8

e) (iii) Strawberry Capacity Too High ($\Delta_S = 1$) - SKIP

Step 1: Check feasibility

$$x_2 = 1 - 1 = 0 \quad \Rightarrow \quad x_2 \text{ leaves the basis}$$

Step 2: Restore optimality

Let t be the new coefficient of x_2 in the Z-row:

	x_1	x_2	x_3	x_4	x_5
$Z(t)$	0	t	0	$0.75 + 0.5t$	$3.5 - t$
x_2	0	1	0	0.5	-1

Step 3: Optimality condition: $3.5 - t \geq 0 \quad \Rightarrow \quad t \leq 3.5$

Step 4: Pivot At $t = 3.5$: x_5 enters, $\quad x_2$ leaves

Result (new shadow prices / reduced costs): $y = (0,\ 2.5,\ 0)$

$\Rightarrow$ Blueberry shadow price increases from 0.75 to 2.5

f) Berry Exchange Optimization

Exchange Δ units between blueberries and strawberries.

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Exchange Δ units between blueberries and strawberries.

From the final tableau: $Z = 41.5 + 0.75\Delta - 3.5\Delta = 41.5 - 2.75\Delta$

	x_1	x_2	x_3	x_4	x_5	Res.
Z	0	0	0	0.75	3.5	41.5
x_3	0	0	1	0	-2	4
x_2	0	1	0	0.5	-1	1
x_1	1	0	0	-0.25	1.5	7.5

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Feasibility:

$$x_3 = 4 + 2\Delta \geq 0, \quad x_2 = 1 + 1.5\Delta \geq 0, \quad x_1 = 7.5 - 1.75\Delta \geq 0$$

Thus:

$$\Delta \in \left[-\frac{2}{3}, \frac{30}{7}\right]$$

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Feasibility:

$$x_3 = 4 + 2\Delta \geq 0, \quad x_2 = 1 + 1.5\Delta \geq 0, \quad x_1 = 7.5 - 1.75\Delta \geq 0$$

Thus:

$$\Delta \in \left[-\frac{2}{3}, \frac{30}{7}\right]$$

Since Z decreases in Δ, choose:

$$\Delta = -\frac{2}{3} \implies Z = 41.5 - 2.75\left(-\frac{2}{3}\right) = 43.33$$

g) New Product: Fruchttraum I

Fruchttraum uses:

$$4 \text{ raspberries}, \quad 2 \text{ blueberries}, \quad 3 \text{ strawberries}$$

Production cost:

$$13$$

Current shadow prices:

$$(0,\ 0.75,\ 3.5)$$

g) New Product: Fruchttraum I

Fruchttraum uses:

4 raspberries, 2 blueberries, 3 strawberries

Production cost:

$$13$$

Current shadow prices:

$$(0,\ 0.75,\ 3.5)$$

Check validity of shadow prices:

- Raspberries: at most 4 available (see optimal tableau) $\Rightarrow$ OK
- Blueberries: valid for decrease ≤ 2 (see e(i)) $\Rightarrow$ OK
- Strawberries: valid for decrease ≤ 5 (see e(ii)) $\Rightarrow$ OK

g) New Product: Fruchttraum I

Fruchttraum uses:

4 raspberries, 2 blueberries, 3 strawberries

Production cost:

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Current shadow prices:

$$(0,\ 0.75,\ 3.5)$$

Check validity of shadow prices:

- Raspberries: at most 4 available (see optimal tableau) $\Rightarrow$ OK
- Blueberries: valid for decrease ≤ 2 (see e(i)) $\Rightarrow$ OK
- Strawberries: valid for decrease ≤ 5 (see e(ii)) $\Rightarrow$ OK

Resource value (opportunity cost):

$$a_{new}^T y = 4 \cdot 0 + 2 \cdot 0.75 + 3 \cdot 3.5 = 12$$

Minimum price:

$$12 + 13 = 25$$

g) New Product: Fruchttraum II

Now Fruchttraum uses:

4 raspberries, 2 blueberries, 6 strawberries

Current strawberry shadow price is valid only for a decrease of 5 units.

For first 5 strawberry units use current shadow prices $(0, 0.75, 3.5)$:

$$4 \cdot 0 + 2 \cdot 0.75 + 5 \cdot 3.5 = 19$$

For the extra 1 strawberry unit, use the new shadow price (e(iii)):

$$1 \cdot 8 = 8$$

Total resource value:

$$19 + 8 = 27$$

Add production cost:

$$27 + 13 = 40 \implies \boxed{\text{Minimum price } = 40}$$

h) Integrality Constraints

If bottles must be produced in integer quantities, add:

$$x_1, x_2 \in \mathbb{Z}_{\geq 0}$$

Then the model becomes an **integer program**.

The LP optimum is:

$$Z^*_{\text{LP}} = 41.5$$

Next IP Modeling Lectures: Since every integer feasible solution is also feasible for the LP relaxation:

$$Z^*_{\text{IP}} \leq Z^*_{\text{LP}}$$

$$\boxed{\text{41.5 is an upper bound for the integer program}}$$

Notes

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